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NIMCET-Free Test



By Amit Katiyar (MCA-JNU)



1. The general solution of $\sqrt{3} \cos x + \sin x = 3$ is

NIMCET 2011

- (a) $2n\pi \pm \frac{\pi}{6}$ (b) $2n\pi \pm \frac{\pi}{3}$
(c) No solution (d) $n\pi \pm \frac{\pi}{6}$

2. The general value of θ satisfying the equation $2 \sin^2 \theta - 3 \sin \theta - 2 = 0$ is

NIMCET 2011

- (a) $n\pi + (-1)^n \frac{\pi}{6}$ (b) $n\pi + (-1)^n \frac{\pi}{2}$
(c) $n\pi + (-1)^n \frac{5\pi}{6}$ (d) $n\pi + (-1)^n \frac{7\pi}{6}$

3. The solution of $\sin x + 1 = \cos x$, such that $0 \leq x \leq 2\pi$ is

NIMCET 2013

- (a) $0, \pi$ (b) $0, \frac{\pi}{2}$
(c) $\frac{\pi}{2}, \frac{3\pi}{2}$ (d) $0, \frac{3\pi}{2}$

4. The value of A that satisfies the equation $a \sin A + b \cos A - c = 0$ is equal to

NIMCET 2017

- (a) $\tan^{-1} \left(\frac{1}{b} \right) \pm \cos^{-1} \left(\frac{c}{\sqrt{a^2+b^2}} \right)$
(b) $\tan^{-1} \left(\frac{c}{b} \right) \pm \sin^{-1} \left(\frac{a}{\sqrt{a^2+b^2}} \right)$
(c) $\tan^{-1} \left(\frac{a}{b} \right) \pm \sin^{-1} \left(\frac{a}{\sqrt{a^2+b^2}} \right)$
(d) None of the above

5. The number of solutions of the equation $\sin x + \sin 5x = \sin 3x$ lying in the interval $[0, \pi]$.

NIMCET 2018

- (a) 4 (b) 6 (c) 5 (d) 2

6. If $\sin^2 x \tan x + \cos^2 x \cos x - \sin 2x = 1 + \tan x + \cot x$, $x \in (0, \pi)$, then x is equal to

NIMCET 2019

- (a) $\frac{3\pi}{12}, \frac{5\pi}{12}$ (b) $\frac{5\pi}{12}, \frac{7\pi}{12}$
(c) $\frac{7\pi}{12}, \frac{11\pi}{12}$ (d) $\frac{7\pi}{12}, \frac{9\pi}{12}$

7. Solve the equation $\sin^2 x - \sin x - 2 = 0$, for x on the interval $0 \leq x \leq 2\pi$.

NIMCET 2020

- (a) $x = -\frac{\pi}{2}$ only (b) $x = \frac{\pi}{4}$ and $\frac{2\pi}{7}$
(c) $x = \frac{2\pi}{3}$ and $\frac{2\pi}{5}$ (d) None of these

8. The general value of θ , satisfying the equation, $\sin \theta = -\frac{1}{2}$, $\tan \theta = \frac{1}{\sqrt{3}}$ is

NIMCET 2021

- (a) $n\pi + \frac{\pi}{6}, n \in I$
(b) $n\pi + (-1)^n \left(\frac{7\pi}{6} \right), n \in I$
(c) $2n\pi + \frac{7\pi}{6}, n \in I$
(d) $2n\pi + \frac{11\pi}{6}, n \in I$

9. If $32 \tan^8 \theta = 2 \cos^2 \alpha - 3 \cos \alpha$ & $3 \cos 2\theta = 1$, then the general value of α for $n \in Z$ is

NIMCET 2021

- (a) $n\pi \pm \frac{\pi}{3}$
(b) $2n\pi \pm \frac{2\pi}{3}$
(c) $2n\pi \pm \frac{\pi}{3}$
(d) $n\pi \pm \frac{2\pi}{3}$

10. The solutions of the equation $4 \cos^2 x + 6 \sin^2 x = 5$ are

NIMCET 2022

- (a) $x = n\pi \pm \frac{\pi}{4}$ (b) $x = n\pi \pm \frac{\pi}{3}$
(c) $x = n\pi \pm \frac{\pi}{2}$ (d) $x = n\pi \pm \frac{2\pi}{3}$

11. The value of $\tan \left(\frac{\pi}{4} + \theta \right) \tan \left(\frac{3\pi}{4} + \theta \right)$ is

- (a) -2 (b) 2
(c) 1 (d) -1

NIMCET 2024

12. If $\sin x = \sin y$ and $\cos x = \cos y$, then the value of $(x - y)$ is (Trigonometry)

- (a) $\pi/4$ (b) $n\pi/2$
(c) $n\pi$ (d) $2n\pi$

NIMCET 2024

13. What is the general solution of the equation $\cot \theta + \tan \theta = 2$?

NIMCET 2025

- (a) $\theta = \frac{n\pi}{2} + \frac{\pi}{4}$
(b) $\theta = \frac{n\pi}{2} + \frac{\pi}{6}$
(c) $\theta = n\pi + \frac{\pi}{4}$
(d) $\theta = \frac{n\pi}{2} + \frac{\pi}{8}$



ANSWER KEY

1.	c	2.	d	3.	d	4.	a	5.	b
6.	c	7.	d	8.	c	9.	b	10.	a
11.	d	12.	d	13.	c				

Analysis 2007 to 2025

Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	00	NIMCET 2016	00
NIMCET 2008	00	NIMCET 2017	01
NIMCET 2009	00	NIMCET 2018	01
NIMCET 2010	00	NIMCET 2019	01
NIMCET 2011	02	NIMCET 2020	01
NIMCET 2012	00	NIMCET 2021	02
NIMCET 2013	01	NIMCET 2022	01
NIMCET 2014	00	NIMCET 2023	00
NIMCET 2015	00	NIMCET 2024	02
		NIMCET 2025	01

SOLUTION

1. (c)

$$2\left(\frac{\sqrt{3}}{2}\cos x + \frac{1}{2}\sin x\right) = 3$$

$$\Rightarrow 2(\sin 60^\circ \cos x + \cos 60^\circ \sin x) = 3$$

$$\Rightarrow \sin(60^\circ + x) = \frac{3}{2}$$

$\Rightarrow \sin \theta$ cannot be greater than 1 ;
hence no solution.

2. (d)

$$2\sin^2 \theta - 3\sin \theta - 2 = 0$$

$$2\sin^2 \theta - 4\sin \theta + \sin \theta - 2 = 0$$

$$\Rightarrow (2\sin \theta + 1)(\sin \theta - 2) = 0$$

$$\Rightarrow \sin \theta = -\frac{1}{2} \text{ or } 2$$

$\therefore \sin \theta \neq 2$

$$\sin \theta = -\frac{1}{2} = \sin\left(-\frac{\pi}{6}\right) = \sin\left(\pi + \frac{\pi}{6}\right)$$

$$\Rightarrow \theta = n\pi + (-1)^n\left(-\frac{\pi}{6}\right)$$

Or $\theta = n\pi + (-1)^n\left(\frac{7\pi}{6}\right)$

3. (d)

$$\sin x + 1 = \cos x, 0 \leq x \leq 2\pi$$

$$\Rightarrow \sin x - \cos x = -1$$

$$\Rightarrow \sqrt{2}\left(\frac{1}{\sqrt{2}}\sin x - \frac{1}{\sqrt{2}}\cos x\right) = -1$$

$$\Rightarrow \sin x \cos 45^\circ - \cos x \sin 45^\circ = \frac{-1}{\sqrt{2}}$$

$$\Rightarrow \sin\left(x - \frac{\pi}{4}\right) = \sin\left(-\frac{\pi}{4}\right)$$

$$= \sin\left(\pi + \frac{\pi}{4}\right) \text{ or } \sin\left(2\pi - \frac{\pi}{4}\right)$$

$$x - \frac{\pi}{4} = -\frac{\pi}{4} \Rightarrow x = 0$$

$$x - \frac{\pi}{4} = \pi + \frac{\pi}{4} \Rightarrow x = \frac{3\pi}{2}$$

And $x - \frac{\pi}{4} = 2\pi - \frac{\pi}{4} \Rightarrow x = 2\pi$

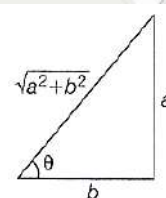
4. (a)

$$a\sin A + b\cos A - c = 0$$

$$\Rightarrow \frac{a}{\sqrt{a^2+b^2}}\sin A + \frac{b}{\sqrt{a^2+b^2}}\cos A - \frac{c}{\sqrt{a^2+b^2}} = 0$$

$$\Rightarrow \sin \theta \sin A + \cos \theta \cos A = \frac{c}{\sqrt{a^2+b^2}}$$

$$\sin \theta = \frac{a}{\sqrt{a^2+b^2}} \text{ and } \cos \theta = \frac{b}{\sqrt{a^2+b^2}}$$



$$\Rightarrow \cos(A - \theta) = \frac{c}{\sqrt{a^2+b^2}}$$

$$\Rightarrow A - \theta = \cos^{-1}\left(\frac{c}{\sqrt{a^2+b^2}}\right)$$

$$\Rightarrow A = \cos^{-1}\left(\frac{c}{\sqrt{a^2+b^2}}\right) + \theta$$

$\therefore \tan \theta = \frac{a}{b} \Rightarrow \theta = \tan^{-1}\left(\frac{a}{b}\right)$

$$\Rightarrow A = \cos^{-1}\left(\frac{c}{\sqrt{a^2+b^2}}\right) + \tan^{-1}\left(\frac{a}{b}\right)$$

Or $\cos(\theta - A) = \frac{c}{\sqrt{a^2+b^2}}$

$$\Rightarrow A = \tan^{-1}\left(\frac{a}{b}\right) - \cos^{-1}\left(\frac{c}{\sqrt{a^2+b^2}}\right)$$

Hence, $A = \tan^{-1}\left(\frac{a}{b}\right) \pm \cos^{-1}\left(\frac{c}{\sqrt{a^2+b^2}}\right)$

5. (b)

$$2\sin 3x \cos 2x - \sin 3x = 0$$

$$\Rightarrow \sin 3x(2\cos 2x - 1) = 0$$

$$\Rightarrow \sin 3x = 0; \cos 2x = \frac{1}{2}$$

$\sin 3x = 0 \Rightarrow 3x = 0, \pi, 2\pi, 3\pi$

Hence, $x = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi$

And $\cos 2x = \frac{1}{2} \Rightarrow 2x = \frac{\pi}{3}, \frac{5\pi}{3}$

Hence, $x = \frac{\pi}{6}, \frac{5\pi}{6} \Rightarrow x = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi \text{ and } \frac{\pi}{6}, \frac{5\pi}{6}$

$\Rightarrow$ Number of solutions = 6.



6. (c)

$$\begin{aligned} \sin^2 x \cdot \tan x + \cos^2 x \cdot \cot x - \sin 2x &= 1 + \tan x + \cot x \\ \Rightarrow \sin^2 x \cdot \tan x + \cos^2 x \cdot \cot x - 2\sin x \cos x &= 1 + \cot x + \tan x \\ \Rightarrow -2\sin x \cos x &= 1 + \tan x(1 - \sin^2 x) + \cot x(1 - \cos^2 x) \\ \Rightarrow -2\sin x \cos x &= 1 + \cos^2 x \tan x + \cot x \cdot \sin^2 x \\ \Rightarrow -2\sin x \cos x &= 1 + \sin x \cos x + \sin x \cos x \\ \Rightarrow \sin 2x &= -\frac{1}{2} \\ \Rightarrow 2x &= \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6} = \frac{7\pi}{6}, \frac{11\pi}{6} \Rightarrow x = \frac{7\pi}{12}, \frac{11\pi}{12} \end{aligned}$$

7. (d) $\sin x = \frac{1 \pm \sqrt{1+8}}{2} = \frac{1 \pm 3}{2} = 2, -1$

$$\because \sin x \neq 2$$

$$\Rightarrow \sin x = -1, x \in [0, 2\pi] \Rightarrow x = \frac{3\pi}{2}$$

8. (c)

$$\sin \theta = -\frac{1}{2} = \sin\left(\pi + \frac{\pi}{6}\right) = \sin \frac{7\pi}{6}$$

$$\Rightarrow \theta = n\pi + (-1)^n \frac{7\pi}{6}, n \in I$$

$$\tan \theta = \frac{1}{\sqrt{3}} = \tan\left(\pi + \frac{\pi}{6}\right) = \tan \frac{7\pi}{6}$$

$$\Rightarrow \theta = n\pi + \frac{7\pi}{6}; n \in I$$

As we know, period of $\sin x$ is 2π and of $\tan x$ is π .

LCM of their periods is 2π .

$$\text{Hence, answer is } \theta = 2n\pi + \frac{7\pi}{6}$$

9. (b)

$$32\tan^8 \theta = 2\cos^2 \alpha - 3\cos \alpha \text{ and } 3\cos 2\theta = 1$$

$$\Rightarrow \cos 2\theta = \frac{1}{3} \Rightarrow 2\cos^2 \theta - 1 = \frac{1}{3}$$

$$\Rightarrow 2\cos^2 \theta = \frac{4}{3} \Rightarrow \cos^2 \theta = \frac{2}{3}$$

$$\Rightarrow \cos \theta = \pm \frac{\sqrt{2}}{\sqrt{3}} \Rightarrow \tan \theta = \pm \frac{1}{\sqrt{2}}$$

$$\Rightarrow \tan^8 \theta = \left(\frac{1}{2^{1/2}}\right)^8 = \frac{1}{2^4} = \frac{1}{16}$$

$$\Rightarrow 32\tan^8 \theta = 2\cos^2 \alpha - 3\cos \alpha$$

$$\Rightarrow 2\cos^2 \alpha - 3\cos \alpha = \frac{32}{16} = 2$$

$$\Rightarrow 2\cos^2 \alpha - 3\cos \alpha - 2 = 0$$

$$\Rightarrow 2\cos^2 \alpha - 4\cos \alpha + \cos \alpha - 2 = 0$$

$$\Rightarrow (2\cos \alpha + 1)(\cos \alpha - 2) = 0 \Rightarrow \cos \alpha = -\frac{1}{2}, 2,$$

$$\because \cos \alpha \neq 2$$

$$\Rightarrow \cos \alpha = -\frac{1}{2} = \cos\left(\pi - \frac{\pi}{3}\right) = \cos\left(\frac{2\pi}{3}\right)$$

$$\Rightarrow \alpha = 2n\pi \pm \frac{2\pi}{3}$$

10. (a)

$$4\cos^2 x + 6\sin^2 x = 5$$

$$\Rightarrow 2\sin^2 x = 1$$

$$\Rightarrow \sin^2 x = \frac{1}{2} = \sin^2 \frac{\pi}{4}$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{4}$$

11. (d) $\tan\left(\frac{\pi}{4} + \theta\right) \tan\left(\frac{3\pi}{4} + \theta\right)$

$$= \tan\left(\frac{\pi}{4} + \theta\right) \tan\left(\frac{\pi}{2} + \frac{\pi}{4} + \theta\right)$$

$$\Rightarrow \tan\left(\frac{\pi}{4} + \theta\right) - \cot\left(\frac{\pi}{4} + \theta\right)$$

$$= -1$$

12. (d)

$\sin x = \sin y$ and $\cos x = \cos y$ then value of $x - y$ is

$$\cos(x-y) = \cos x \cdot \cos y + \sin x \cdot \sin y$$

$$= \cos x \cdot \cos x + \sin x \cdot \sin x$$

$$= \cos^2 x + \sin^2 x$$

$$\cos(x-y) = 1$$

$$x - y = 2n\pi$$

13. (c)

Given:

$$\cot \theta + \tan \theta = 2$$

$$\text{We use the identity: } \cot \theta = \frac{1}{\tan \theta}$$

$$\frac{1}{\tan \theta} + \tan \theta = 2$$

Let $x = \tan \theta$, then:

$$x + \frac{1}{x} = 2$$

$$x^2 - 2x + 1 = 0$$

$$(x - 1)^2 = 0$$

$$x = 1 \Rightarrow \tan \theta = 1$$

So:

$$\tan \theta = 1 \Rightarrow \theta = n\pi + \frac{1}{4}, n \in \mathbb{Z}$$

$$\text{Ans. (c) } \theta = n\pi + \frac{\pi}{4}$$